LET'S LEARN

PERCENTAGES

A Complete Step-by-Step Lesson

■ **Level**	Middle School & Above
■ **Duration**	90 – 120 minutes
■ **Topics**	Concept • Types • Formulas • Applications
■ **Includes**	Examples • Practice • Word Problems • Quiz

TABLE OF CONTENTS

1	What Is a Percentage?
2	Percentages on a Number Line & Visualisation
3	Converting: Percentages, Fractions & Decimals
4	The Three Core Percentage Formulas
5	Finding a Percentage of a Quantity
6	Finding What Percentage One Number Is of Another
7	Finding the Whole from a Percentage
8	Percentage Increase & Decrease
9	Profit, Loss & Discount
10	Simple Interest
11	Worked Examples
12	Practice Problems & Quiz

1 | What Is a Percentage?

■ **Definition**

The word "percentage" comes from the Latin "per centum" meaning per hundred.
A percentage expresses a number as a fraction of 100.
We use the symbol % to denote a percentage.
So 45% means 45 out of every 100.

Why Do We Use Percentages?

Percentages give us a universal, easy-to-compare way of expressing proportions. Whether we are talking about exam scores, discounts in a shop, rainfall statistics or nutritional information on food packaging, percentages appear everywhere in daily life.

■ **School**	You scored 36 out of 40 → 90%
■ **Shopping**	30% OFF the marked price
■ **Finance**	8% annual interest rate
■ **Statistics**	65% of voters preferred Option A
■ **Nutrition**	Fat content: 12% of daily value

2 | Visualising Percentages

Key Benchmarks to Remember

Knowing these landmark percentages helps you estimate quickly:

Percentage	Fraction	Decimal	What it means
1%	1/100	0.01	One in every hundred
10%	1/10	0.10	One in every ten
20%	1/5	0.20	One in every five
25%	1/4	0.25	One quarter
33.3%	1/3	0.33…	One third
50%	1/2	0.50	One half
75%	3/4	0.75	Three quarters
100%	1/1	1.00	The whole amount
200%	2/1	2.00	Double the amount

■ Memory Tip

Think of a 100-square grid. Each small square = 1%.
Colour 50 squares → 50% is shaded.
Drawing or imagining such a grid is a powerful mental model.

3 | Converting Between Percentages, Fractions & Decimals

Conversion	Method	Example
% → Fraction	Divide by 100, then simplify	35% = 35/100 = 7/20
% → Decimal	Divide by 100 (move point left 2)	35% = 0.35
Fraction → %	Multiply by 100	3/4 × 100 = 75%
Decimal → %	Multiply by 100 (move point right 2)	0.6 × 100 = 60%

Worked Conversion Examples

■ **Convert 68% to a fraction**

Step: 68% = 68/100

→ *Simplify by dividing by 4: = 17/25*

■ **Convert 0.375 to a percentage**

Step: 0.375 × 100 = 37.5%

■ **Convert 5/8 to a percentage**

Step: 5 ÷ 8 = 0.625

→ *0.625 × 100 = 62.5%*

■ **Convert 125% to a decimal**

Step: 125 ÷ 100 = 1.25

→ *(Greater than 1 → more than the whole)*

4 | The Three Core Percentage Formulas

Every percentage calculation is a variation of the same triangle of three quantities: Part, Whole, and Percentage. Know any two and you can find the third.

Find the PART	**Part = (Percentage ÷ 100) × Whole**	*What is 30% of 200? = (30/100) × 200 = 60*
Find the %	**Percentage = (Part ÷ Whole) × 100**	*15 out of 60 as %? = (15/60) × 100 = 25%*
Find the WHOLE	**Whole = (Part ÷ Percentage) × 100**	*18 is 45% of what? = (18/45) × 100 = 40*

■ The Percentage Triangle

Draw a triangle with PART at the top, WHOLE and % at the bottom.
Cover the value you want to find:
- Cover PART → Whole × (% ÷ 100)
- Cover WHOLE → Part ÷ (% ÷ 100)
- Cover % → (Part ÷ Whole) × 100

5 | Finding a Percentage of a Quantity

Part = (Percentage ÷ 100) × Whole

Step-by-Step Method

Step 1: Write down the percentage as a fraction over 100.

Step 2: Multiply that fraction by the whole quantity.

Step 3: Simplify / calculate the result.

■ Example: Find 15% of 80

$15\% \text{ of } 80 = (15/100) \times 80$

$= 0.15 \times 80$

$= 12$

■ Example: A jacket costs ■2,400. VAT is 18%. Find the VAT amount.

$\text{VAT} = 18\% \text{ of } 2400$

$= (18/100) \times 2400$

$= 0.18 \times 2400$

= ■432

■ Example: A school has 850 students. 40% are girls. How many girls?

$\text{Girls} = 40\% \text{ of } 850$

$= (40/100) \times 850$

$= 340 \text{ girls}$

6 | What Percentage Is One Number of Another?

Percentage = (Part ÷ Whole) × 100

Examples

- **Example: Express 24 as a percentage of 96.**

  = (24 ÷ 96) × 100

  = 0.25 × 100

  = 25%

- **Example: A student scored 54 out of 90 in a test. What is the percentage score?**

  = (54 ÷ 90) × 100

  = 0.6 × 100

  = 60%

- **Example: Out of 500 items, 35 were defective. What % were defective?**

  = (35 ÷ 500) × 100

  = 0.07 × 100

  = 7%

7 | Finding the Whole from a Percentage

$$\text{Whole} = (\text{Part} \div \text{Percentage}) \times 100$$

■ Example: 18 is 25% of what number?

Whole $= (18 \div 25) \times 100$

$= 0.72 \times 100$

$= 72$

■ Example: A shop reduced a price by ■360, which was 15% off. What was the original price?

Whole $= (360 \div 15) \times 100$

$= 24 \times 100$

= ■2,400

■ Example: After paying 30% tax, Anita received ■4,900. What was her original salary?

Anita received 70% of her salary.

Whole $= (4900 \div 70) \times 100$

$= 70 \times 100$

= ■7,000

8 | Percentage Increase & Decrease

	Formula	Quick Multiplier
Increase by P%	New = Old + (P/100 × Old) = Old × (1 + P/100)	Multiply by (100+P)/100
Decrease by P%	New = Old – (P/100 × Old) = Old × (1 – P/100)	Multiply by (100–P)/100

Percentage Change Formula

% Change = [(New – Old) ÷ Old] × 100

■ **Example: A shirt costs ■800. Its price increases by 12.5%. New price?**

New price = 800 × (1 + 12.5/100)
= 800 × 1.125
= ■900

■ **Example: Population of a town was 40,000. After a year it is 46,000. % increase?**

% Increase = [(46000 – 40000) / 40000] × 100
= (6000 / 40000) × 100
= 15%

■ **Example: A car was bought for ■6,00,000 and sold for ■4,80,000. % decrease?**

% Decrease = [(600000 – 480000) / 600000] × 100
= (120000 / 600000) × 100
= 20%

9 | Profit, Loss & Discount

Term	Formula
Cost Price (CP)	Price paid to buy / produce the item
Selling Price (SP)	Price at which item is sold
Profit	SP – CP (when SP > CP)
Loss	CP – SP (when CP > SP)
Profit %	(Profit / CP) × 100
Loss %	(Loss / CP) × 100
Marked Price (MP)	Price listed / advertised
Discount	MP – SP
Discount %	(Discount / MP) × 100
SP after discount	SP = MP × (1 – Discount%/100)

■ Example: A shopkeeper buys a radio for ■1,200 and sells it for ■1,500. Find profit %.

Profit = 1500 – 1200 = ■300

Profit% = (300 / 1200) × 100 = 25%

■ Example: An item with MP ■800 is sold at 15% discount. Find SP.

SP = 800 × (1 – 15/100)

= 800 × 0.85 = ■680

10 | Simple Interest

$$SI = (P \times R \times T) \div 100$$

Symbol	Meaning
P	Principal — the original amount of money
R	Rate — the annual interest rate (%)
T	Time — in years
SI	Simple Interest earned
A	Total Amount = P + SI

■ Example: Find SI on ■5,000 at 8% p.a. for 3 years.

SI = (5000 × 8 × 3) / 100

= 1,20,000 / 100

SI = ■1,200

Total Amount = 5000 + 1200 = ■6,200

■ Example: In how many years will ■2,400 earn ■576 as SI at 6% per annum?

576 = (2400 × 6 × T) / 100

576 = 144T

T = 576 / 144 = 4 years

11 | Fully Worked Examples

These multi-step problems combine concepts from across the lesson.

Problem 1 — Two Successive Discounts

A TV has a marked price of ■24,000. It is sold with successive discounts of 20% and 10%. Find the selling price.

After 1st discount (20%): SP1 = 24,000 × 0.80 = ■19,200

After 2nd discount (10%): SP2 = 19,200 × 0.90 = ■17,280

Final Selling Price = ■17,280

Note: The combined discount is NOT 30% — it is only 28%.

Problem 2 — Population Growth Over Two Years

A town had a population of 50,000. It grew 10% in year 1 and 5% in year 2. Find the population after two years.

After year 1: 50,000 × 1.10 = 55,000

After year 2: 55,000 × 1.05 = 57,750

Final population = 57,750

Problem 3 — Finding Cost Price from Profit%

A seller makes 25% profit and sells an item for ■6,250. What was the cost price?

SP = CP × (1 + 25/100) = CP × 1.25

CP = 6250 / 1.25 = ■5,000

12 | Practice Problems & Quiz

Section A — Short Answer (1–2 marks each)

Q1. Convert 7/20 into a percentage.

Q2. What is 35% of 260?

Q3. Express 48 as a percentage of 160.

Q4. The price of milk increased from ■40 to ■46 per litre. Find the % increase.

Q5. Find the whole if 72 is 60% of it.

Q6. Convert 0.045 to a percentage.

Q7. A coat costs ■3,500 and is reduced by 30%. Find the new price.

Section B — Word Problems (3–4 marks each)

Q8. A car is bought for ■5,50,000 and sold at a loss of 8%. What is the selling price?

Q9. Ravi scored 78%, 82%, 91% and 69% in four tests each out of 100. Find his overall percentage.

Q10. A sum of ■8,000 is invested at 7.5% per annum simple interest for 4 years. Find the total amount at the end.

Q11. An article is marked at ■1,600 and sold at a 12.5% discount. If the cost price is ■1,200, find the profit percentage on CP.

Q12. A factory increased production by 15% in January and then decreased by 10% in February from the January level. If the February output is 20,700 units, what was the original (December) output?

Answer Key — Section A

1. 35%	2. 91
3. 30%	4. 15%
5. 120	6. 4.5%
7. ■2,450	

■ Section B — Answers (for teacher)

Q8: ■5,06,000

Q9: 80%

Q10: ■10,400

Q11: 16.67%

Q12: 20,000 units

Quick-Reference Formula Card

Concept	Formula
Part from %	Part = (% ÷ 100) × Whole
% of a Whole	% = (Part ÷ Whole) × 100
Whole from %	Whole = (Part ÷ %) × 100
% Increase/Decrease	% Change = [(New – Old) ÷ Old] × 100
New value (increase)	New = Old × (1 + %/100)
New value (decrease)	New = Old × (1 – %/100)
Profit %	(Profit ÷ CP) × 100
Loss %	(Loss ÷ CP) × 100
Discount %	(Discount ÷ MP) × 100
Simple Interest	SI = (P × R × T) ÷ 100
Total Amount	A = P + SI

Well done! You've completed the Percentages lesson. Practise these formulas daily and you'll master percentages in no time! ■